Impossible plans can't be saved.

Costly plans need a name on them.

One live picture of the portfolio. An agent that proposes re-plans.
Planners who decide. Built on the Databricks lakehouse.

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THE PROBLEM AS WE UNDERSTAND IT

One question takes days to answer, and margin depends on it

Margin is fragile

- Single-digit operating margins
- Penalties capped around 5 to 10% of contract
- One slipped milestone can erase a programme's profit

Risk spreads silently

- A few rigs and crews serve 20 to 25 programmes
- One rig down moves risk to other programmes
- Nobody sees it until a date is missed

Planning lives in heads

- Excel and local factory tools
- Constraint knowledge in a few senior planners
- Key-person risk on the core of the business

The question nobody can answer in less than days today

“If this bid lands, or this test rig goes down, what happens to everything else, and to our margin?”

WHAT CHANGES

From reconciling spreadsheets to deciding on one live picture

	Today	With the system
Portfolio view	Assembled by hand, weeks old	Live: every programme, milestone and scarce resource
Rig and crew bookings	Trackers per rig and per crew	One plan of record. A double booking cannot be saved
“What if this bid lands?”	Days of planner work	A scenario in minutes, with its penalty price
A rig goes down	Phone calls, re-plan over days	A costed re-plan waiting the same day; the planner decides
Who decided what	Email threads	Every commit versioned: who, why, at what cost

THE HEART OF THE DESIGN

Five rules the plan cannot break

R1	**One campaign per rig**	and only on a rig that can run that test
R2	**Qualified crews only**	certified for that vehicle, on those dates
R3	**Crews free and inside the window**	travel time included, inside the customer's night or weekend access
R4	**Halls have a fixed number of bays**	parked car-bodies never exceed the physical space
R5	**Every plan carries its price**	late milestones, penalties, what is recoverable. Costly plans need a director

R1 to R4 block what is physically impossible. R5 prices what is merely expensive. Each is a database rule, tested, not a guideline: it binds planners and the agent alike.

HOW IT STAYS SAFE

The agent proposes. The database refuses what is impossible. A person decides.

Rig A breaks down

10-week outage recorded. Two test campaigns now conflict.

Agent proposes

Moves both campaigns to other capable rigs.
Price: type approval 59 days late, EUR 2.95M.

Planner improves it

Pulls another campaign forward.
Price down to EUR 1.4M, 28 days late.

Director signs

Above the penalty threshold (R5), so a director must approve.

Committed as plan v2

One transaction, versioned, conflicts cleared.

Refused along the way, by the database (17 executable tests)

Attempt	Rule	Who
Book a rig that cannot run brake tests; double-book Rig B	R1	Agent
Send a crew not qualified for the metro	R2	Agent
Test outside the customer's window; same crew with no travel time	R3	Agent
Commit the cheaper plan without a director	R5	Planner

REFUSALS ARE DATA

Every refused change is kept, with the rule that refused it

Audit

- Who tried what, when
- Which rule stopped it
- Planner or agent, same log

Agent learning

- Each refusal is feedback for the next try
- History becomes its evaluation set
- Shows which rule it misreads

Capacity decisions

- Pressure per rig, crew pool, hall
- Sustained refusals justify renting or hiring
- No pressure avoids buying

Resource	Refusals in the test run	Rules	Reading
RIG-B	4	R1	The only free EMC rig: the bottleneck after an outage
CRW-1	2	R3	One qualified metro crew; windows are tight
Hall H1, bay 1	1	R4	Parking pressure while batches wait

BUSINESS VALUE

Protect margin first, then grow the order book safely

Lever	How	Order of magnitude / year
Avoid penalties	Risk visible weeks earlier, same-day re-plans	EUR 5 to 10M
Recover from disruptions	~2 weeks recovered per major disruption	EUR 2 to 4M (partly overlaps)
Bid on dates we can meet	Bids checked against real rig and crew capacity	Avoid EUR 70M tail risk on one bid
Capacity on evidence	Refusal pressure shows where to rent or hire	Better-timed capex
Planner capacity	Scenarios in minutes; knowledge captured	Resilience, less key-person risk

How to read it

- Assumes EUR 7.7bn in flight, 6% margin, penalties ~1% of revenue
- Penalty lever alone ≈ 3 to 7% of operating profit
- Board-level case: bid risk and resilience
- Build ≈ EUR 1 to 1.5M, payback < 1 year if the penalty assumption holds

First number to get in week 1: penalties actually paid or accrued in the last 3 years, by milestone. If it is 0.3% of revenue, not 1%, the case rests on bid risk and resilience.

THE ORDER WE UNLOCK IT

Each step pays for itself and makes the next one safe. The agent comes last.

1 · See

Weeks 0–8

- Sources in, IDs resolved
- Portfolio risk picture
- Read-only, low risk

2 · Commit safely

Weeks 6–16

- Rig, crew, hall bookings
- R1 to R5 enforced
- Trackers retired

3 · What-if

Weeks 12–20

- Bid and disruption scenarios
- Price per option
- Better bid decisions

4 · Agent proposes

Weeks 18–26

- Rig-down re-plan first
- Planner approves or edits
- Days to hours

Where we push back

The ask is an agent on top. On today's data it would reason over spreadsheets that disagree. Steps 1 and 2 deliver most of the risk reduction and make the agent safe. If we stopped after step 2, Ferronova would already have one trusted plan.

WHAT MAKES OR BREAKS IT

Three things decide success. Only one of them is technical.

1 · Planners want to switch

- If the app is slower than the lead planner's Excel, it dies
- They see their own rules in the model
- They keep the final say

2 · One write path

- Every rig, crew and hall change goes through the plan of record
- Trackers retired, not kept in parallel
- A management decision

3 · Data you can trust

- Factories name things differently
- ID crosswalk with named owners
- Penalty terms as data, not contract PDFs

OPEN QUESTIONS

What I need from you to firm this up

Business

- What did penalties cost in the last 3 years, by milestone?
- Which constraint binds most today: rigs, crews or hall space?
- How are penalties written: caps, recovery, force majeure for rig failure?
- Who may accept a plan that makes a customer milestone late?
- At what bid probability should a bid reserve capacity?
- How much autonomy for the agent in year one?

Technical

- Direct DB access with change capture on the scheduling tool?
- Which MES per factory, and what can each export, how often?
- Does the register cover batches, rigs and crews, or only programmes?
- Hall space: bays, or metres?
- Lakebase in your region, with the extensions we need?
- How fresh must the picture be: minutes, hours, daily?

HOW I WORKED WITH AI

Faster on the mechanical parts, overruled on the judgment calls

Where it moved me faster

- [Replace with your own example]
- Drafting DDL, seed data and a test harness
- Time saved went into deciding which rules block and which only price

Where I overrode it

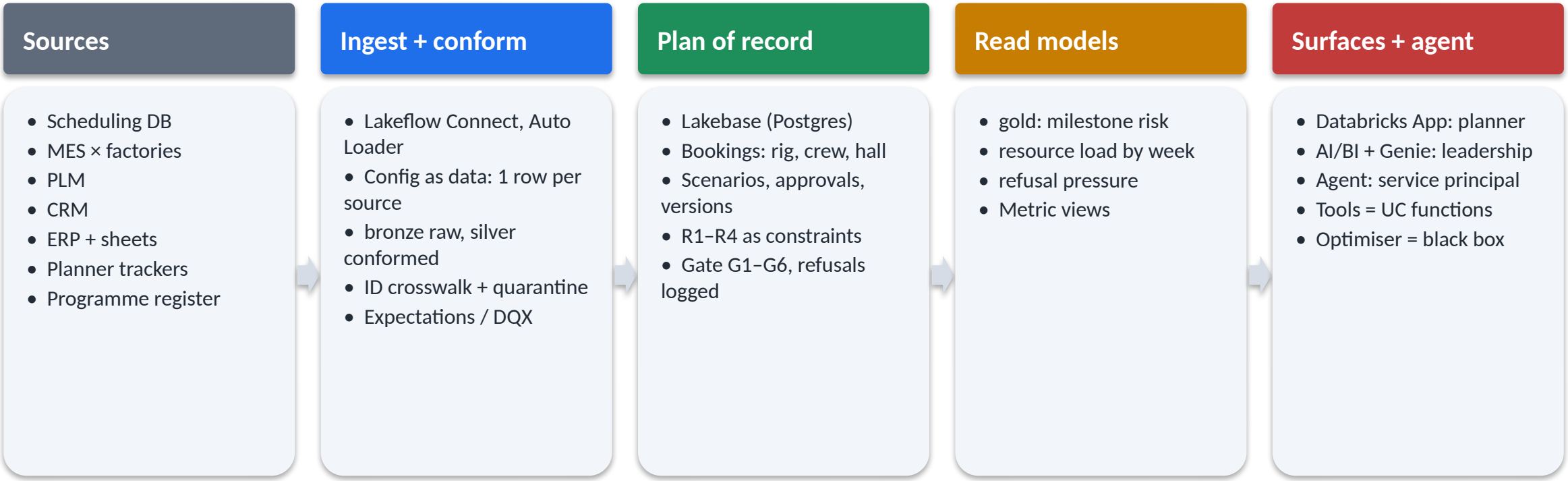
- [Replace with your own example]
- Checks after the fact detect a double booking; they cannot refuse it
- Agent safety moved out of the prompt, into permissions and constraints

Adapt this slide to your own session before sending.

Backup · architecture, data model, agent, planner prototype

BACKUP · ARCHITECTURE

Lakehouse for the picture, Lakebase for the plan of record



- Why Lakebase: Delta enforces only row-level rules. “One campaign per rig at a time” is a cross-row rule. Postgres makes it impossible to save.
- Data quality (is this data trustworthy?) lives in the pipelines. Decision validity (is this plan committable?) lives only at commit.
- Decisions do not round-trip through bronze and silver: the ledger is their system of record, replicated to Delta for history.

BACKUP · DATA MODEL

Every rule maps to a database object, scoped by scenario

Rule	Mechanism
R1 one campaign per rig	EXCLUDE (scenario, rig, period &&) + FK rig_capability
R2 qualified crew	FK (crew, design, qual_period) + CHECK period <@ qual_period
R3 in window, crew free	FK (window, site, window_period) + CHECK; EXCLUDE (crew, [mobilise, end))
R4 hall never over-committed	Bays as positions + EXCLUDE (scenario, position, period &&)
R5 every plan priced	v_penalty_exposure; G5 escalates above threshold
No commit on outages	Gate G4 over v_plan_conflicts
No stale overwrite	G2: base_version check under lock on plan_head
Refusals kept	rule_catalog + rejection_log + v_rejection_pressure

Core tables

- Reference: programme, design, design_type_test, milestone, campaign
- Resources: test_rig, rig_capability, crew, crew_qualification, access_window, hall_position
- Plan (by scenario): rig_booking, crew_assignment, hall_occupancy
- Governance: scenario, approval, plan_version, operational_event, guardrail_policy, rejection_log

Scoped to one trigger first: a test rig goes down

Its whole action list

- branch_scenario: open its own draft
- try_move_booking: move a campaign
- try_assign_crew: place a crew
- submit_scenario: hand over with rationale
- Each returns OK or the refusing rule

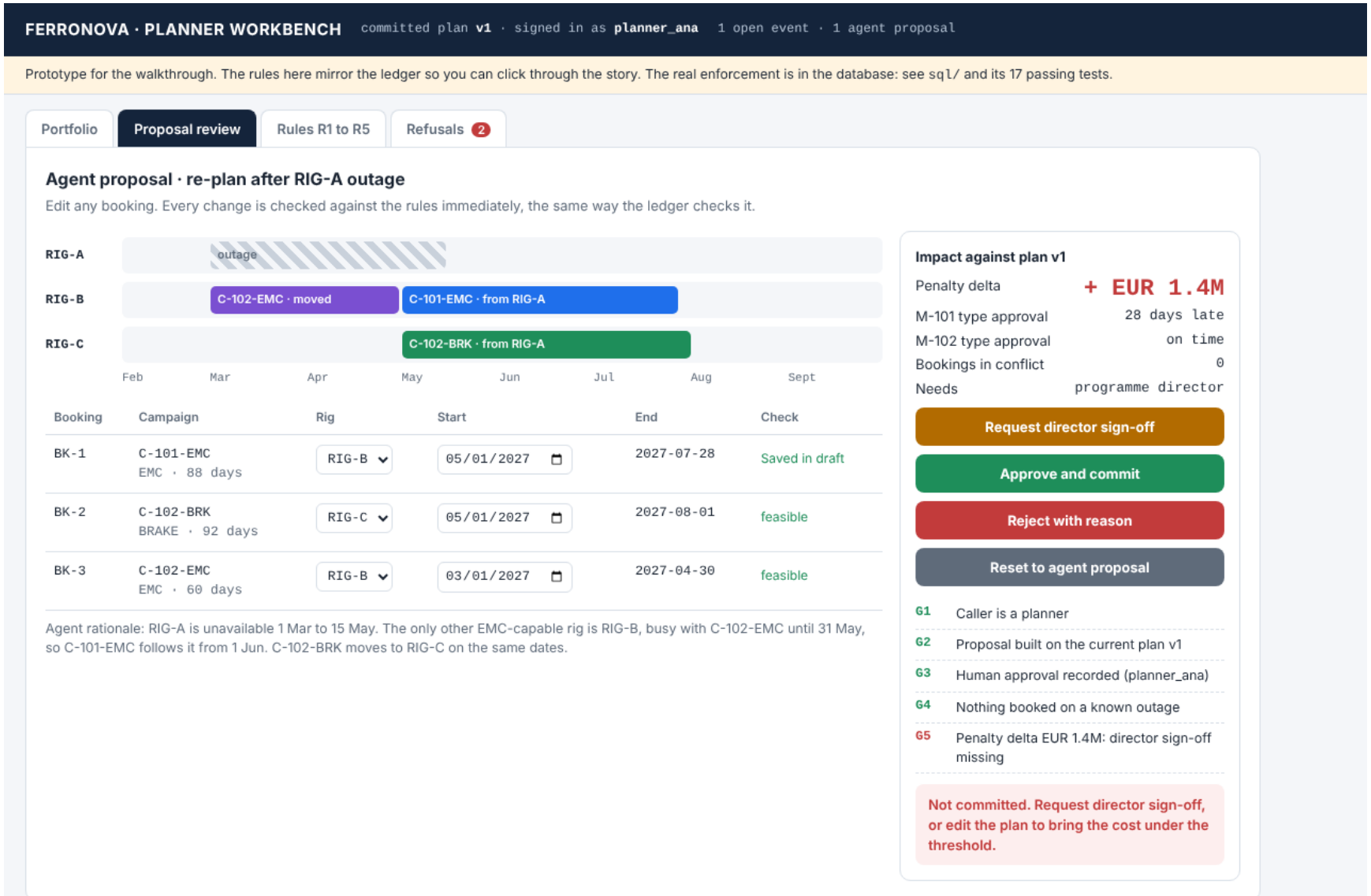
Never touches

- Contract dates and penalty terms
- Customer windows, qualifications, rig capabilities
- Factory shop-floor sequencing
- The committed plan, approvals, thresholds
- Bid pricing and customer talks

Guardrails live in the database

- RLS: own draft only
- Constraints: every row feasible
- No permission to commit
- G2 stale, G4 reality, G5 cost → director
- MLflow trace + refusal log

planner-prototype.html: the proposal, the price, and the gate



Try it live

- Agent proposal: EUR 2.95M
- Move C-101 to 1 May: refused R1
- Pull C-102-EMC to 1 Mar, then C-101 to 1 May: EUR 1.4M
- Commit: G5 asks for a director
- Director signs: plan v2